

Problem 1

Uniform distribution. A random variable that is equally likely to take any value in a finite set S is said to have the uniform distribution on S . If U is such a random variable and $\emptyset \neq R \subseteq S$, show that the distribution of U conditional on $\{U \in R\}$ is uniform on R .

Proof. Since U is uniformly distributed on S , we have: □

Problem 2

3. Let X be a random variable with distribution function

$$\mathbb{P}\{X \leq x\} = \begin{cases} 0 & \text{if } x \leq 0, \\ x & \text{if } 0 < x \leq 1, \\ 1 & \text{if } x > 1. \end{cases}$$

Let F be a distribution function which is continuous and strictly increasing. Show that $Y = F^{-1}(X)$ is a random variable having distribution function F .

Is it necessary that F be continuous and/or strictly increasing?

Proof. □

Problem 3

(a) Show that any discrete random variable may be written as a linear combination of indicator variables.

(b) Show that any random variable may be expressed as the limit of an increasing sequence of discrete random variables.

(c) Show that the limit of any increasing convergent sequence of random variables is a random variable.

Proof. □

Problem 4

Express the distribution functions of

$$X^+ = \max\{0, X\}, \quad X^- = -\min\{0, X\}, \quad |X| = X^+ + X^-, \quad -X,$$

in terms of the distribution function F of the random variable X .

Proof.

□

Problem 5

The real number m is called a median of the distribution function F whenever $\lim_{y \rightarrow m} F(y) \leq \frac{1}{2} \leq F(m)$.

(a) Show that every distribution function F has at least one median, and that the set of medians of F is a closed interval of $\mathbb{R}$.

(b) Show, if F is continuous, that $F(m) = \frac{1}{2}$ for any median m .

Proof.

□

Problem 6

Let $(\Omega, \mathcal{F})$ be a measurable space, $A_1, \dots, A_n \in \mathcal{F}$ and $b_1, \dots, b_n \in \mathbb{R}$. Set $X = b_1 1_{A_1} + \dots + b_n 1_{A_n}$. Suppose that for all $i \neq j$, $A_i \cap A_j = \emptyset$ and $b_i \neq b_j$. Show that $\sigma(X) = \sigma(\{A_1, \dots, A_n\})$.

Proof.

□

Problem 7

Consider the random walk on $\{0, 1, \dots, N\}$ that reflects at 0, introduced in the class. More precisely, let $S_0 = k$, $S_{j+1} = S_j + X_j$, where $X_j = 1$ if $S_j = 0$, otherwise $\mathbb{P}[X_j = 1] = p$ and $\mathbb{P}[X_j = -1] = q$. Compute the expected number of steps to first reach state N , as a function of k . Distinguish the cases $p = q$ and $p \neq q$.

Proof.

□

Problem 8

If $F : \mathbb{R} \rightarrow [0, 1]$ satisfies monotone increasing, right continuity, $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$, show that F is the distribution function of some random variable on some probability space. (Hint: you may want to revisit Section 2.3, exercise 3)

Proof.

□